AMARNATH S. PATEL

Orlando, FL | amarnath.patel@ucf.edu | apatel.co | github.com/jeebuscrossaint | ORCID 0009-0008-9460-082X

Optical engineering — interferometric benches, fiber and laser instrumentation, metasurface design and tolerancing.

EDUCATION

University of Central Florida — **Undergraduate Student** — **4.00 GPA**
B.S. Physics (Optics & Lasers) and Mathematics, Computer Science minor — August 2025 – Expected May 2029

- **Optics and physics:** Geometric Optics & Lab, Modern Physics, Quantum Information Processing, Mathematical Methods for Physics, Electricity & Magnetism I (in progress), Independent Research [PHY 4912]
- Optics coursework through CREOL, UCF's College of Optics and Photonics.

Florida Atlantic University — **University coursework via FAU High School, ages 14–18** — **3.66 GPA**
111 credit hours completed between ages 14 and 18, alongside high school — August 2021 – May 2025

OPTICS EXPERIENCE

Undergraduate Researcher — UCF Astrophotonics Lab, CREOL — August 2025 – Present
Supervisor: Dr. Stephen Eikenberry

- Align and operate an off-axis digital holography bench to measure the wavelength-dependent complex transfer matrix of photonic lanterns — the amplitude and phase relationship between each single-mode input port and the multimode output — across the C-band, 1525–1575 nm, extending Dobias et al., *Opt. Express* **34**(9), 17217 (2026).
- Hands-on with the full optical chain: interferometer alignment and fringe optimization, fiber coupling and switching, tunable IR source control, InGaAs imaging, and motorized polarization control. Instruments include an HP 8168E tunable laser, a Xenics Bobcat 320 InGaAs camera, a DiCon GP700 fiber switch and a Thorlabs MPC320 polarization controller.
- Implemented the phase-retrieval and mode-decomposition analysis: FFT sideband isolation and demodulation, Butterworth low-pass filtering, joint optimization of mode-field diameter, defocus quadratic phase and field position, then LP-basis decomposition for complex modal amplitude and phase.
- Delivered the per-port reconstruction-fidelity characterization for 6- and 7-port lanterns at ≈ 98% fidelity against simulated fields — the metrology result the group's measurement campaign rests on.
- Automated the four-instrument acquisition chain over GPIB/VISA, RS-232 and GigE Vision so a complete all-port × C-band sweep runs unattended, with polarization optimized in-loop for peak fringe contrast and saturated frames rejected on arrival. A full dataset went from days of bench time to a single hands-off run.
- Own the telescope-node software for PolyOculus, an instrument program building large-aperture observing capability from a networked array of 8 telescopes fed by photonic linkage to a stabilized spectrograph: autoguiding from a DMK 33UX273 guide camera to iOptron mount guide pulses over INDI, holding a star on a target pixel such as the fiber (simulated tracking error 3.01″ → 0.29″ RMS at 0.254″/px), and closed-loop piezo fast-steering-mirror control holding a laser spot to 0.2 px. Brought the nodes' Jetson Nanos up on mainline Linux to replace NVIDIA's end-of-life image.

Independent Research — CELERIS: metasurface design and analysis pipeline — 2026

- Built a rigorous coupled-wave analysis solver from scratch (1D TE/TM gratings with Li and Liu–Fan inverse-rule factorization, full 2D-vectorial formulation for biperiodic nanopillar arrays, Redheffer scattering-matrix recursion) and the complete metalens design pipeline on top of it.
- Optical design and analysis: Sellmeier and tabulated n, k dispersion, meta-atom library sweeps with automatic etch-depth selection for full-2π phase coverage, gradient-based inverse design, and Pancharatnam–Berry geometric-phase and polarization-multiplexed lenses exploiting form birefringence, reported as per-polarization RMS phase and focal isolation in dB.
- Performance evaluated the way an optical system is: Strehl ratio, Zernike wavefront decomposition, MTF, chromatic focal shift, through-focus caustic, and Monte-Carlo fabrication-tolerance analysis — with fabrication-ready GDSII export at the end of it.
- Validated against closed-form physics, an independent method and energy conservation (1e-6 to 1e-15), cross-checked against the grcwa and Stanford S⁴ solvers (≈ 1e-7), and reproduces canonical published metalenses (Khorasaninejad et al., *Science* 2016; Chen et al., *Nat. Nanotechnol.* 2018). Diffraction-limited designed focal spots at FWHM = $\lambda f/D$. *Sole-author manuscript in preparation.*

TECHNICAL SKILLS

Optics and metrology: interferometric alignment, off-axis digital holography, phase retrieval, modal decomposition, wavefront analysis (Strehl, Zernike, MTF), fiber coupling and photonic lanterns, polarization control, diffractive and metasurface design, tolerance analysis

Instrumentation: GPIB/VISA and RS-232 instrument control, GigE Vision and UVC camera acquisition, INDI mount control and autoguiding, piezo fast steering mirrors, tunable IR lasers, InGaAs imaging, motorized stages and polarization controllers, FITS and GDSII

Computation: electromagnetic simulation (RCWA/FMM), FFT and signal processing, gradient-based optimization, Monte-Carlo methods, GPU/CUDA acceleration; C/C++ (C++23), CUDA, Python (NumPy, SciPy), Rust; Linux, Git, CMake, Qt/PySide6

HONORS

Florida Bright Futures — Florida Academic Scholars (highest tier; 100% tuition) — 2025
Lockheed Martin Award, "Highest Level of Engineering Excellence," AEV Competition — 2024